

2026 GOSPEL AWARDS

FOH CHANNEL PROCESSING — DiGiCo Quantum 225

Input List Summary

Ch	Instrument	Mic / DI
1	Kick	Earthworks DM6
2	Snare	Lauten Snare Mic (LS-408)
3	Underhat	Audix M1280BHC
4	Rack 1	Earthworks DM17
5	Rack 2	Earthworks DM17
6	Floor Tom	Earthworks DM17
7	Underhead Left	Audix M1280B
8	Underhead Right	Audix M1280B
9	Bass DI	Radial Pro DI
10	Guitar Dynamic	AE2500 — Dynamic Element
11	Guitar Condenser	AE2500 — Condenser Element
12	Keys 1 Left	Radial Pro DI
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14	Keys 2 Left	Radial Pro DI
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16	Tracks	Radial Pro DI
17	Background Vocal 1	SM58
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Ch 1 – Kick — Earthworks DM6

Section	Parameter / Value	Details
HPF	40 Hz @ 24 dB/Oct	
LPF	8 kHz @ 12 dB/Oct	
Band1; High Shelf	-3 dB @ 6 kHz	Q 0.8; Shelf — Tame condenser edge; punchy, not brittle
Band2; Upper-Mid (Param)	+4 dB @ 3 kHz	Q 1.5 — Beater click and attack; drives kick through a dense gospel arrangement
Band3; Lower-Mid (Param/Dyn)	-6 dB @ 250 Hz	Q 2.0; Dyn On; Thr -14 dB; A 8 ms; R 60 ms — Memo standing wave; tightens as drummer pushes harder
Band4; Low (Lowshelf)	+4 dB @ 60 Hz	Q 1.2; Shelf — Sub fundamental weight and punch
Dynamics 1; Compressor	On	Thr -8 dB Ratio 6:1 A 5 ms R 80 ms Gain +4 dB Knee Hard
Dynamics 2; Gate	On	Thr -40 dB A 3 ms H 30 ms R 150 ms Range 70 dB
Quick Summary		DM6's wide condenser response captures the full kick spectrum. Low shelf builds sub punch. Dynamic EQ at 250 Hz targets the Memo's primary standing wave — compounds as drummer pushes. Beater click at 3 kHz drives kick through the arrangement. Gate at -40 dB sits comfortably above stage bleed (~-50 dBFS) with headamp gain at nominal.

Ch 2 – Snare — Lauten Snare Mic (LS-408)

Section	Parameter / Value	Details
HPF	100 Hz @ 24 dB/Oct	Set LS-408 onboard HP switch to 80 Hz; console adds the extra slope
LPF	Off	Leave LS-408 onboard LP flat
Band1; High Shelf	+2 dB @ 10 kHz	Q 0.8; Shelf — Air and shimmer; LS-408 handles the extension cleanly
Band2; Upper-Mid (Param)	+4 dB @ 5 kHz	Q 1.2 — Snap and crack definition
Band3; Lower-Mid (Param/Dyn)	-4 dB @ 250 Hz	Q 2.0; Dyn On; Thr -16 dB; A 8 ms; R 80 ms — Memo standing wave; tightens dynamically as hits intensify
Band4; Low (Param)	+2 dB @ 160 Hz	Q 1.5 — Shell body; just below the 200 Hz Memo standing wave problem
Dynamics 1; Compressor	On	Thr -10 dB Ratio 4:1 A 6 ms R 80 ms Gain +3 dB Knee Hard
Dynamics 2; Gate	On	Thr -38 dB A 2 ms H 20 ms R 100 ms Range 70 dB
Quick Summary		LS-408 28 dB off-axis rejection is already doing heavy lifting in a loud gospel context. Body at 160 Hz stays below the Memo 200 Hz problem. Dynamic EQ at 250 Hz tightens automatically on harder hits. Gate at -38 dB opens cleanly on hits without false-triggering on kick or tom bleed at nominal gain.

Ch 3 – Underhat — Audix M1280BHC

Section	Parameter / Value	Details
Placement	Below hat	2-4 inches below bottom cymbal, angled up at 30-45 deg. Mount on goose-neck off hat stand or short boom. Keep out of camera sightline — this is the entire point.
HPF	200 Hz @ 24 dB/Oct	Lower than top-position: underside captures more body and fundamental, so HPF moves down; still need to clear kick bleed
LPF	Off	
Band1; High Shelf	+4 dB @ 12 kHz	Q 0.8; Shelf — Underside pickup is inherently darker; heavier HF lift needed vs top position
Band2; Upper-Mid (Param)	+4 dB @ 6 kHz	Q 1.2 — Compensate for reduced stick-attack transient; underside misses the top-plate impact — add presence here
Band3; Lower-Mid (Param)	-3 dB @ 600 Hz	Q 2.0 — Metallic clank from underside; similar to top position
Band4; Low (Param)	-5 dB @ 315 Hz	Q 1.5 — Memo standing wave PLUS kick/shell bleed are more prominent in underhat position; heavier cut than top-position
Dynamics 1; Compressor	On	Thr -18 dB Ratio 2:1 A 20 ms R 200 ms Gain +1 dB Knee Soft
Dynamics 2; Gate	On	Thr -42 dB A 3 ms H 10 ms R 60 ms Range 60 dB — Tighter than top-position; kick bleed is heavier from below
Polarity	CHECK	Check polarity against kick in mono. Underhat position frequently inverts vs close mics depending on placement angle.
Quick Summary		Underhat position (broadcast visual restriction) points mic upward at the underside of the bottom cymbal. Result: inherently darker pickup with more body, less top-plate transient. Lower HPF to 200 Hz to capture that body. The two heaviest adjustments vs top-position: more HF boost (the underside lacks the stick-impact shimmer of the top plate) and a heavier 315 Hz cut (kick bleed and shell resonance are more prominent from below). Gate tightens to -42 dB for the same reason. Polarity check is not optional — flip and compare in mono.

Ch 4 – Rack 1 — Earthworks DM17

Section	Parameter / Value	Details
HPF	100 Hz @ 24 dB/Oct	
LPF	12 kHz @ 12 dB/Oct	
Band1; High Shelf	-2 dB @ 8 kHz	Q 0.8; Shelf — Tame condenser edge; toms warm, not brittle
Band2; Upper-Mid (Param)	+3 dB @ 4 kHz	Q 1.2 — Attack and smack
Band3; Lower-Mid (Param/Dyn)	-5 dB @ 315 Hz	Q 2.0; Dyn On; Thr -14 dB; A 8 ms; R 100 ms — Memo standing wave; prevents boom buildup on hard hits
Band4; Low (Param)	+3 dB @ 200 Hz	Q 1.5 — Tom shell body; balance against the Memo standing wave at this freq
Dynamics 1; Compressor	On	Thr -10 dB Ratio 4:1 A 8 ms R 120 ms Gain +3 dB Knee Medium
Dynamics 2; Gate	On	Thr -42 dB A 4 ms H 20 ms R 150 ms Range 70 dB
Quick Summary		DM17 rim-mount gives consistent placement. Body at 200 Hz with dynamic cut at 315 Hz balances warmth against the Memo two worst standing wave frequencies. Gate at -42 dB gives clean separation between hits and inter-drum bleed at nominal headamp gain. Adjust threshold at soundcheck if tom hits are not tracking cleanly.

Ch 5 – Rack 2 — Earthworks DM17

Section	Parameter / Value	Details
HPF	100 Hz @ 24 dB/Oct	
LPF	12 kHz @ 12 dB/Oct	
Band1; High Shelf	-2 dB @ 8 kHz	Q 0.8; Shelf — Tame condenser edge
Band2; Upper-Mid (Param)	+3 dB @ 4 kHz	Q 1.2 — Attack and definition
Band3; Lower-Mid (Param/Dyn)	-5 dB @ 315 Hz	Q 2.0; Dyn On; Thr -14 dB; A 8 ms; R 100 ms — Memo standing wave; boom control on hits
Band4; Low (Param)	+3 dB @ 200 Hz	Q 1.5 — Tom shell body; shift to match this drum tuning
Dynamics 1; Compressor	On	Thr -10 dB Ratio 4:1 A 8 ms R 120 ms Gain +3 dB Knee Medium
Dynamics 2; Gate	On	Thr -42 dB A 4 ms H 20 ms R 150 ms Range 70 dB
Quick Summary		Match Rack 1 as starting point. Shift Band 1 frequency to match this drum tuning. Gate at -42 dB matches Rack 1 — adjust both rack gates relative to each other for consistent response across fills.

Ch 6 – Floor Tom — Earthworks DM17

Section	Parameter / Value	Details
HPF	60 Hz @ 24 dB/Oct	Lower than rack toms — floor fundamental lives at 60–100 Hz
LPF	10 kHz @ 12 dB/Oct	
Band1; High Shelf	-3 dB @ 8 kHz	Q 0.8; Shelf — Keep it warm and deep, not bright
Band2; Upper-Mid (Param)	+2 dB @ 3 kHz	Q 1.2 — Stick attack
Band3; Lower-Mid (Param/Dyn)	-5 dB @ 250 Hz	Q 2.0; Dyn On; Thr -12 dB; A 8 ms; R 100 ms — Memo problem zone; floor toms resonate hard here
Band4; Low (Lowshelf)	+4 dB @ 100 Hz	Q 1.2; Shelf — Deep floor tom thump and weight
Dynamics 1; Compressor	On	Thr -8 dB Ratio 4:1 A 8 ms R 150 ms Gain +3 dB Knee Medium
Dynamics 2; Gate	On	Thr -45 dB A 4 ms H 30 ms R 200 ms Range 70 dB
Quick Summary		Lower HPF to 60 Hz and build the low shelf aggressively — floor tom fundamentals live down here. 250 Hz is a prime Memo standing wave; floor toms resonate hard there. Gate at -45 dB with slower release — do not chop the tail on a floor hit in a quiet passage.

Ch 7 – Underhead Left — Audix M1280B

Section	Parameter / Value	Details
Placement	Below cymbals	12-18 inches below outermost cymbal undersides, angled up 30-45 deg. Position OUTSIDE the shell footprint to minimize shell resonance pickup. Short boom stands or goose-necks off cymbal stands.
HPF	200 Hz @ 24 dB/Oct	Same as overhead — still want to exclude shell body; close mics handle that
LPF	Off	
Band1; High Shelf	+4 dB @ 12 kHz	Q 0.8; Shelf — Heavier than overhead; underheads are inherently darker; more air boost needed
Band2; Upper-Mid (Param)	+4 dB @ 8 kHz	Q 1.0 — Heavier than overhead; underside pickup lacks top-plate shimmer — compensate here
Band3; Lower-Mid (Param/Dyn)	-5 dB @ 315 Hz	Q 2.0; Dyn On; Thr -16 dB; A 20 ms; R 200 ms — Heavier than overhead; shell resonance at 315 Hz more prominent from below
Band4; Low (Lowshelf)	-5 dB @ 200 Hz	Q 2.0; Shelf — HEAVIER than overhead: mics now sit in the shell zone; low-mid mud and Memo standing wave accumulate more from this position
Dynamics 1; Compressor	On	Thr -16 dB Ratio 2:1 A 25 ms R 300 ms Gain +2 dB Knee Soft
Dynamics 2; Ducker	On	Thr -50 dB A 10 ms H 50 ms R 300 ms Range 40 dB — Light ducker; not a hard gate
Polarity	CHECK FIRST	Underheads frequently require polarity flip vs close mics. Check Ch 7 and Ch 8 together against Ch 1 (kick) and Ch 2 (snare) in mono before anything else. Should sound fuller — if thinner, flip polarity on both underheads.
Reverb Send	Increase +2 to +3 dB	Underheads capture less natural room air than overheads; compensate by increasing reverb bus send on Ch 7/8
Quick Summary		Underhead position (broadcast visual restriction) — mics below cymbals angled up. Inherently darker and less transient than overhead. Two key adjustments vs overhead starting points: heavier low-mid cuts (-5 dB at both 200 and 315 Hz) since mics sit inside the shell zone, and heavier HF boost (+4 dB at 8 kHz and 12 kHz shelf) to compensate for the loss of top-plate shimmer. POLARITY: check Ch 7/8 against close mics in mono before soundcheck — this is not optional. Increase reverb send +2 to 3 dB vs normal overhead send to restore natural air. Link Ch 7 and Ch 8 as stereo pair on Q225.

Ch 8 – Underhead Right — Audix M1280B

Section	Parameter / Value	Details
Placement	Below cymbals	Match Ch 7 — same height, same angle. Mirror image placement, outside shell footprint on stage right.
HPF	200 Hz @ 24 dB/Oct	
LPF	Off	
Band1; High Shelf	+4 dB @ 12 kHz	Q 0.8; Shelf — Match Ch 7
Band2; Upper-Mid (Param)	+4 dB @ 8 kHz	Q 1.0 — Match Ch 7
Band3; Lower-Mid (Param/Dyn)	-5 dB @ 315 Hz	Q 2.0; Dyn On; Thr -16 dB; A 20 ms; R 200 ms — Match Ch 7
Band4; Low (Lowshelf)	-5 dB @ 200 Hz	Q 2.0; Shelf — Match Ch 7
Dynamics 1; Compressor	On	Thr -16 dB Ratio 2:1 A 25 ms R 300 ms Gain +2 dB Knee Soft
Dynamics 2; Ducker	On	Thr -50 dB A 10 ms H 50 ms R 300 ms Range 40 dB
Polarity	CHECK FIRST	Flip polarity both Ch 7 and Ch 8 together (not independently). Check in mono against close mics. Adjust as a matched pair.
Reverb Send	Increase +2 to +3 dB	Match Ch 7 — increase to compensate for reduced natural room air from underhead position
Quick Summary		Match Ch 7 exactly — link as a stereo pair on the Q225. Polarity must be checked and matched on both channels simultaneously. Mirror placement of Ch 7 on stage right. Verify stereo image during soundcheck; nudge angle or panning if kit sounds asymmetric.

Ch 9 – Bass DI — Radial Pro DI

Section	Parameter / Value	Details
HPF	40 Hz @ 24 dB/Oct	
LPF	5 kHz @ 12 dB/Oct	Bass DI does not need top end
Band1; High Shelf	+2 dB @ 3.5 kHz	Q 0.8; Shelf — String click and pick attack
Band2; Upper-Mid (Param)	+3 dB @ 1 kHz	Q 1.5 — Midrange growl and definition; audibility across all PA zones
Band3; Lower-Mid (Param/Dyn)	-5 dB @ 250 Hz	Q 2.0; Dyn On; Thr -16 dB; A 10 ms; R 100 ms — Memo worst standing wave; bass accumulates here fast
Band4; Low (Lowshelf)	+3 dB @ 80 Hz	Q 1.2; Shelf — Fundamental weight and punch
Dynamics 1; Compressor	On	Thr -16 dB Ratio 4:1 A 8 ms R 100 ms Gain +3 dB Knee Medium
Dynamics 2; Gate	Off	No gate on bass — sustain is intentional
Quick Summary		Passive Pro DI is clean and transparent. Low shelf adds fundamental without bumping what the PA subs are already handling. 250 Hz is the worst Memo standing wave on bass. Comp threshold at -16 dB reflects the continuous DI signal at nominal gain (crest factor ~8-10 dB puts peaks around -15 to -12 dBFS). No gate.

Ch 10 – Guitar Dynamic — AE2500 — Dynamic Element

Section	Parameter / Value	Details
HPF	80 Hz @ 24 dB/Oct	
LPF	6 kHz @ 12 dB/Oct	Crossover point — condenser element (Ch 11) handles above here
Band1; High Shelf	-4 dB @ 4 kHz	Q 0.8; Shelf — Roll off where condenser element (Ch 11) takes over
Band2; Upper-Mid (Param)	+3 dB @ 1.5 kHz	Q 1.5 — Midrange grit and presence
Band3; Lower-Mid (Param/Dyn)	-5 dB @ 315 Hz	Q 2.0; Dyn On; Thr -16 dB; A 10 ms; R 80 ms — Memo problem zone; cab boom that builds as player drives harder
Band4; Low (Param)	+2 dB @ 150 Hz	Q 1.5 — Cabinet warmth and body
Dynamics 1; Compressor	On	Thr -14 dB Ratio 4:1 A 10 ms R 100 ms Gain +2 dB Knee Medium
Dynamics 2; Gate	On	Thr -55 dB A 5 ms H 20 ms R 150 ms Range 70 dB
Quick Summary		AE2500 dynamic element captures low-mid warmth and body. LPF at 6 kHz creates a crossover — blend with Ch 11. Comp at -14 dB catches peaks on a driven guitar amp. Gate at -55 dB — guitar signal drops near the noise floor when amp is idle; this keeps it clean without cutting off amp breath. Match gate timing with Ch 11.

Ch 11 – Guitar Condenser — AE2500 — Condenser Element

Section	Parameter / Value	Details
HPF	250 Hz @ 24 dB/Oct	Crossover from dynamic element — condenser handles above here
LPF	Off	
Band1; High Shelf	+2 dB @ 8 kHz	Q 0.8; Shelf — Air and top-end definition
Band2; Upper-Mid (Param)	+3 dB @ 3.5 kHz	Q 1.2 — Presence and cut in a dense gospel arrangement
Band3; Lower-Mid (Param/Dyn)	-4 dB @ 400 Hz	Q 2.0; Dyn On; Thr -18 dB; A 15 ms; R 100 ms — Condenser cab boxiness below the crossover
Band4; Low (Lowshelf)	-4 dB @ 250 Hz	Q 1.5; Shelf — Crossover cleanup; prevent frequency stacking with Ch 10
Dynamics 1; Compressor	On	Thr -14 dB Ratio 3:1 A 15 ms R 120 ms Gain +2 dB Knee Medium
Dynamics 2; Gate	On	Thr -55 dB A 5 ms H 20 ms R 150 ms Range 70 dB
Quick Summary		Condenser element handles top-end character and transient detail. HPF at 250 Hz creates a clean crossover with Ch 10. Blend: start condenser ~3–4 dB below dynamic, adjust by ear. Match gate timing with Ch 10. Both elements need 48V.

Ch 12 – Keys 1 Left — Radial Pro DI

Section	Parameter / Value	Details
HPF	80 Hz @ 24 dB/Oct	
LPF	Off	Keys span the full frequency range; resist heavy LPF filtering
Band1; High Shelf	+1 dB @ 8 kHz	Q 0.8; Shelf — Top-end sparkle
Band2; Upper-Mid (Param)	+2 dB @ 3 kHz	Q 1.2 — Presence and clarity
Band3; Lower-Mid (Param/Dyn)	-4 dB @ 315 Hz	Q 2.0; Dyn On; Thr -16 dB; A 15 ms; R 150 ms — Second standing wave; varies with playing intensity
Band4; Low (Lowshelf)	-3 dB @ 200 Hz	Q 1.5; Shelf — Memo standing wave; DI keys honk badly here in a reverberant room
Dynamics 1; Compressor	On	Thr -16 dB Ratio 3:1 A 15 ms R 150 ms Gain +2 dB Knee Medium
Dynamics 2; Gate	Off	No gate on keys — release tails are musical
Quick Summary		Resist heavy LPF — keys span the full range. Comp at -16 dB reflects continuous DI source at nominal gain. Both low-mid cuts address Memo standing waves that pile up fast on DI keyboard sources. Link Ch 12 and 13 as stereo pair. Monitor for midrange stacking against Keys 2.

Ch 13 – Keys 1 Right — Radial Pro DI

Section	Parameter / Value	Details
HPF	80 Hz @ 24 dB/Oct	
LPF	Off	
Band1; High Shelf	+1 dB @ 8 kHz	Q 0.8; Shelf — Match Ch 12
Band2; Upper-Mid (Param)	+2 dB @ 3 kHz	Q 1.2 — Match Ch 12
Band3; Lower-Mid (Param/Dyn)	-4 dB @ 315 Hz	Q 2.0; Dyn On; Thr -16 dB; A 15 ms; R 150 ms — Match Ch 12
Band4; Low (Lowshelf)	-3 dB @ 200 Hz	Q 1.5; Shelf — Match Ch 12
Dynamics 1; Compressor	On	Thr -16 dB Ratio 3:1 A 15 ms R 150 ms Gain +2 dB Knee Medium
Dynamics 2; Gate	Off	
Quick Summary		Link to Ch 12 as stereo pair — match all settings, adjust both simultaneously. Confirm signal is actually stereo before panning wide.

Ch 14 – Keys 2 Left — Radial Pro DI

Section	Parameter / Value	Details
HPF	80 Hz @ 24 dB/Oct	
LPF	Off	
Band1; High Shelf	+1 dB @ 8 kHz	Q 0.8; Shelf — Air
Band2; Upper-Mid (Param)	+2 dB @ 3 kHz	Q 1.2 — Presence
Band3; Lower-Mid (Param/Dyn)	-4 dB @ 315 Hz	Q 2.0; Dyn On; Thr -16 dB; A 15 ms; R 150 ms — Standing wave control
Band4; Low (Lowshelf)	-3 dB @ 200 Hz	Q 1.5; Shelf — Memo standing wave
Dynamics 1; Compressor	On	Thr -16 dB Ratio 3:1 A 15 ms R 150 ms Gain +2 dB Knee Medium
Dynamics 2; Gate	Off	
Quick Summary		Match Keys 1 template. Link Ch 14 and 15 as stereo pair. If Keys 2 runs at a different level, adjust comp threshold accordingly. Carve midrange between Keys 1 and Keys 2 if they stack.

Ch 15 – Keys 2 Right — Radial Pro DI

Section	Parameter / Value	Details
HPF	80 Hz @ 24 dB/Oct	
LPF	Off	
Band1; High Shelf	+1 dB @ 8 kHz	Q 0.8; Shelf — Match Ch 14
Band2; Upper-Mid (Param)	+2 dB @ 3 kHz	Q 1.2 — Match Ch 14
Band3; Lower-Mid (Param/Dyn)	-4 dB @ 315 Hz	Q 2.0; Dyn On; Thr -16 dB; A 15 ms; R 150 ms — Match Ch 14
Band4; Low (Lowshelf)	-3 dB @ 200 Hz	Q 1.5; Shelf — Match Ch 14
Dynamics 1; Compressor	On	Thr -16 dB Ratio 3:1 A 15 ms R 150 ms Gain +2 dB Knee Medium
Dynamics 2; Gate	Off	
Quick Summary		Link to Ch 14 as stereo pair — match all settings exactly.

Ch 16 – Tracks — Radial Pro DI

Section	Parameter / Value	Details
HPF	80 Hz @ 24 dB/Oct	
LPF	Off	
Band1; High Shelf	Flat	
Band2; Upper-Mid (Param)	Flat	
Band3; Lower-Mid (Param)	-3 dB @ 315 Hz	Q 2.0 — Second standing wave; static cut (tracks run at consistent level)
Band4; Low (Lowshelf)	-2 dB @ 200 Hz	Q 1.5; Shelf — Memo standing wave on playback source
Dynamics 1; Compressor	On	Thr -18 dB Ratio 2:1 A 20 ms R 200 ms Gain +1 dB Knee Soft — Level consistency only
Dynamics 2; Gate	Off	
Quick Summary		Tone shaping belongs to the track producer — this is level management only. Comp at -18 dB is appropriate for a consistent playback source that does not vary in level like a live instrument. Standing wave cuts still apply; reverberant rooms accumulate those frequencies on any source. Confirm track content with production before show.

Ch 17 – Background Vocal 1 — SM58

Section	Parameter / Value	Details
HPF	100 Hz @ 24 dB/Oct	
LPF	15 kHz @ 12 dB/Oct	
Band1; High Shelf	+3 dB @ 8 kHz	Q 0.7; Shelf — Adds air; compensates for SM58 roll-off above 7 kHz
Band2; Upper-Mid (Param)	+4 dB @ 3 kHz	Q 1.2 — Intelligibility and presence
Band3; Lower-Mid (Param/Dyn)	-5 dB @ 300 Hz	Q 2.0; Dyn On; Thr -16 dB; A 10 ms; R 80 ms — Boxy buildup; six open BG mics compound this aggressively in a reverberant room
Band4; Low (Lowshelf)	-4 dB @ 200 Hz	Q 1.5; Shelf — Proximity effect + Memo standing wave — one cut handles both
Dynamics 1; Compressor	On	Thr -14 dB Ratio 4:1 A 12 ms R 100 ms Gain +3 dB Knee Medium
Mustard; Processing	On	Mustard: Purple (Optical) Thr -12 dB Ratio 3:1 A 15 ms R 120 ms Gain +2 dB — Smooth vintage leveling
Dynamics 2; Gate	On	Thr -40 dB A 5 ms H 20 ms R 100 ms Range 70 dB
Quick Summary		Six SM58s open simultaneously creates compounding problems at 300 Hz in a 1.6-second room. Comp at -14 dB catches vocal peaks (~-14 to -10 dBFS at nominal gain). Mustard threshold at -12 dB adds smooth optical leveling above that. Gate at -40 dB sits between the singing level (RMS -25 to -20) and stage bleed floor (~-50 dBFS). Start all six channels identical — adjust gate per singer at soundcheck.

Ch 18 – Background Vocal 2 — SM58

Section	Parameter / Value	Details
HPF	100 Hz @ 24 dB/Oct	
LPF	15 kHz @ 12 dB/Oct	
Band1; High Shelf	+3 dB @ 8 kHz	Q 0.7; Shelf — Match Ch 17
Band2; Upper-Mid (Param)	+4 dB @ 3 kHz	Q 1.2 — Match Ch 17
Band3; Lower-Mid (Param/Dyn)	-5 dB @ 300 Hz	Q 2.0; Dyn On; Thr -16 dB; A 10 ms; R 80 ms — Match Ch 17
Band4; Low (Lowshelf)	-4 dB @ 200 Hz	Q 1.5; Shelf — Match Ch 17
Dynamics 1; Compressor	On	Thr -14 dB Ratio 4:1 A 12 ms R 100 ms Gain +3 dB Knee Medium
Mustard; Processing	On	Mustard: Purple (Optical) Thr -12 dB Ratio 3:1 A 15 ms R 120 ms Gain +2 dB
Dynamics 2; Gate	On	Thr -40 dB A 5 ms H 20 ms R 100 ms Range 70 dB — Adjust per singer
Quick Summary		Match Ch 17. Adjust gate threshold per singer level at soundcheck. Keep EQ and comp identical across all six BG channels so the section blends as a unit.

Ch 19 – Background Vocal 3 — SM58

Section	Parameter / Value	Details
HPF	100 Hz @ 24 dB/Oct	
LPF	15 kHz @ 12 dB/Oct	
Band1; High Shelf	+3 dB @ 8 kHz	Q 0.7; Shelf — Match Ch 17
Band2; Upper-Mid (Param)	+4 dB @ 3 kHz	Q 1.2 — Match Ch 17
Band3; Lower-Mid (Param/Dyn)	-5 dB @ 300 Hz	Q 2.0; Dyn On; Thr -16 dB; A 10 ms; R 80 ms — Match Ch 17
Band4; Low (Lowshelf)	-4 dB @ 200 Hz	Q 1.5; Shelf — Match Ch 17
Dynamics 1; Compressor	On	Thr -14 dB Ratio 4:1 A 12 ms R 100 ms Gain +3 dB Knee Medium
Mustard; Processing	On	Mustard: Purple (Optical) Thr -12 dB Ratio 3:1 A 15 ms R 120 ms Gain +2 dB
Dynamics 2; Gate	On	Thr -40 dB A 5 ms H 20 ms R 100 ms Range 70 dB — Adjust per singer
Quick Summary		Match Ch 17. Adjust gate threshold per singer at soundcheck.

Ch 20 – Background Vocal 4 — SM58

Section	Parameter / Value	Details
HPF	100 Hz @ 24 dB/Oct	
LPF	15 kHz @ 12 dB/Oct	
Band1; High Shelf	+3 dB @ 8 kHz	Q 0.7; Shelf — Match Ch 17
Band2; Upper-Mid (Param)	+4 dB @ 3 kHz	Q 1.2 — Match Ch 17
Band3; Lower-Mid (Param/Dyn)	-5 dB @ 300 Hz	Q 2.0; Dyn On; Thr -16 dB; A 10 ms; R 80 ms — Match Ch 17
Band4; Low (Lowshelf)	-4 dB @ 200 Hz	Q 1.5; Shelf — Match Ch 17
Dynamics 1; Compressor	On	Thr -14 dB Ratio 4:1 A 12 ms R 100 ms Gain +3 dB Knee Medium
Mustard; Processing	On	Mustard: Purple (Optical) Thr -12 dB Ratio 3:1 A 15 ms R 120 ms Gain +2 dB
Dynamics 2; Gate	On	Thr -40 dB A 5 ms H 20 ms R 100 ms Range 70 dB — Adjust per singer
Quick Summary		Match Ch 17. Adjust gate threshold per singer at soundcheck.

Ch 21 – Background Vocal 5 — SM58

Section	Parameter / Value	Details
HPF	100 Hz @ 24 dB/Oct	
LPF	15 kHz @ 12 dB/Oct	
Band1; High Shelf	+3 dB @ 8 kHz	Q 0.7; Shelf — Match Ch 17
Band2; Upper-Mid (Param)	+4 dB @ 3 kHz	Q 1.2 — Match Ch 17
Band3; Lower-Mid (Param/Dyn)	-5 dB @ 300 Hz	Q 2.0; Dyn On; Thr -16 dB; A 10 ms; R 80 ms — Match Ch 17
Band4; Low (Lowshelf)	-4 dB @ 200 Hz	Q 1.5; Shelf — Match Ch 17
Dynamics 1; Compressor	On	Thr -14 dB Ratio 4:1 A 12 ms R 100 ms Gain +3 dB Knee Medium
Mustard; Processing	On	Mustard: Purple (Optical) Thr -12 dB Ratio 3:1 A 15 ms R 120 ms Gain +2 dB
Dynamics 2; Gate	On	Thr -40 dB A 5 ms H 20 ms R 100 ms Range 70 dB — Adjust per singer
Quick Summary		Match Ch 17. Adjust gate threshold per singer at soundcheck.

Ch 22 – Background Vocal 6 — SM58

Section	Parameter / Value	Details
HPF	100 Hz @ 24 dB/Oct	
LPF	15 kHz @ 12 dB/Oct	
Band1; High Shelf	+3 dB @ 8 kHz	Q 0.7; Shelf — Match Ch 17
Band2; Upper-Mid (Param)	+4 dB @ 3 kHz	Q 1.2 — Match Ch 17
Band3; Lower-Mid (Param/Dyn)	-5 dB @ 300 Hz	Q 2.0; Dyn On; Thr -16 dB; A 10 ms; R 80 ms — Match Ch 17
Band4; Low (Lowshelf)	-4 dB @ 200 Hz	Q 1.5; Shelf — Match Ch 17
Dynamics 1; Compressor	On	Thr -14 dB Ratio 4:1 A 12 ms R 100 ms Gain +3 dB Knee Medium
Mustard; Processing	On	Mustard: Purple (Optical) Thr -12 dB Ratio 3:1 A 15 ms R 120 ms Gain +2 dB
Dynamics 2; Gate	On	Thr -40 dB A 5 ms H 20 ms R 100 ms Range 70 dB — Adjust per singer
Quick Summary		Match Ch 17. Adjust gate threshold per singer at soundcheck.

Ch 24 – Podium — Earthworks Podium Mic

Section	Parameter / Value	Details
HPF	120 Hz @ 24 dB/Oct	Speech; nothing useful below this
LPF	12 kHz @ 12 dB/Oct	Limit to speech range; reduces feedback risk
Band1; High Shelf	+2 dB @ 6 kHz	Q 0.8; Shelf — Consonant clarity and definition
Band2; Upper-Mid (Param)	+4 dB @ 2.5 kHz	Q 1.2 — Speech intelligibility — vowel formants and consonant articulation
Band3; Lower-Mid (Param/Dyn)	-5 dB @ 315 Hz	Q 2.0; Dyn On; Thr -18 dB; A 10 ms; R 100 ms — Second standing wave; dynamic catches varying speaker distances from podium
Band4; Low (Lowshelf)	-5 dB @ 200 Hz	Q 1.5; Shelf — Memo standing wave; clearing this is critical for speech intelligibility
Dynamics 1; Compressor	On	Thr -16 dB Ratio 4:1 A 10 ms R 150 ms Gain +3 dB Knee Medium — Level variation between presenters
Dynamics 2; Ducker	On	Thr -50 dB A 8 ms H 50 ms R 400 ms Range 40 dB — Gentle ducker; not a hard gate
Quick Summary		Earthworks podium mic is the cleanest condenser in the rig — EQ is almost entirely Memo standing wave removal. Comp at -16 dB handles level variation between presenters at a continuous-signal nominal gain. Gentle ducker (not a hard gate) — a hard gate will cut off quiet or distant voices. Ducker at -50 dB opens well above ambient noise; slow release prevents pumping.

GLOBAL REVERB — SEVENTH HEAVEN PRO (WET MODE)

Usage	Preset / Settings
BG Vocal Reverb — Main Hall	Preset: Warm Hall Decay 1.4s • PreDelay 22 ms • Early/Late 40/60 • HF Damp 3.5 • Mix 12% — Pull decay back from preset default; the Memo is adding 1.6s of real room. Sweetening on return: HS +2 dB @ 8 kHz; LC 120 Hz
BG Vocal Reverb — Tight Plate	Preset: Gold Plate Decay 0.9s • PreDelay 15 ms • Early/Late 45/55 • HF Damp 3.8 • Mix 10% — Warm, dense plate for blend and intimacy. Sweetening on return: HS +1 dB @ 10 kHz; LC 100 Hz
Drum Room	Preset: Drum Room Decay 0.5s • PreDelay 6 ms • Early/Late 55/45 • HF Damp 4.2 • Mix 8% — Short and punchy; use primarily on snare send. Sweetening on return: LC 150 Hz
Keys Reverb	Preset: Concert Hall Decay 1.2s • PreDelay 28 ms • Early/Late 35/65 • HF Damp 3.0 • Mix 14% — Pull decay well below preset default; room is doing the grand hall work already. Sweetening on return: HS +2 dB @ 10 kHz; LC 80 Hz
Overall Glue	Preset: Live Room Decay 0.7s • PreDelay 6 ms • Early/Late 50/50 • HF Damp 3.8 • Mix 6% — Subtle cohesion across the mix bus or master sends. Sweetening on return: LC 200 Hz

CRITICAL NOTES — REVERBERANT ROOM (RT60 ≈ 1.6s):

- All reverb mix % assume 100% wet (send/return). Adjust channel sends as needed.
- Use LESS reverb than normal — the room adds 1.6 seconds of natural reverb. Start conservative.
- Pay close attention to LC filters on reverb returns — essential to prevent low-mid mud buildup.
- Dynamic EQ settings are calibrated for a live reverberant environment. Trust the room.
- Memo problem frequencies (200 Hz, 250–315 Hz) accumulate on every open mic — treat them all.
- All settings are starting points. Trust your ears and adjust during soundcheck.

WORKFLOW REMINDERS:

- Set gains FIRST — target RMS -25 to -20 dBFS at headamp before applying any EQ or dynamics.
- Gate thresholds are calibrated for that gain structure. Verify at soundcheck before tweaking.
- AE2500 (Ch 10/11): Both elements need 48V. Blend on a bus or VCA — start condenser 3-4 dB below dynamic.
- POLARITY CHECK — Underhat (Ch 3) and Underheads (Ch 7/8): Check polarity against close drum mics in mono BEFORE soundcheck. Underheads frequently require polarity flip as a matched pair.
- Link stereo pairs: Underheads (Ch 7/8) • Keys 1 (Ch 12/13) • Keys 2 (Ch 14/15).
- VCA groups: Kit (Ch 1–8) • Band DI (Ch 9, 12–16) • Guitar (Ch 10–11) • BG Vocals (Ch 17–22) • Podium (Ch 24) — Note: Ch 3 = Underhat, Ch 7/8 = Underheads (broadcast visual restriction)
- Crowd mic rig (OM1, Deity S2, CM4) — patch as standard every show. EQ starting points in venue-notes.md.